Neuromorphic X2

512 x 512 ReRAM Compute-in-Memory Macro

32-bit AHB Subordinate Interface
Pin Description and Integration Rules

Property	Value
Model	Neuromorphic_X2_ahb_beh
Default register address	0xE000_0004
AHB data width	32 bits
Crossbar	512 rows x 512 columns
Command / response FIFOs	32 entries / 32 entries
Document revision	2.0 - 30 October 2026

The behavioral interface is an AHB subordinate and is compatible with a standards-compliant 32-bit AHB master when the address, transfer-size, and handshake requirements in this document are followed.

Physical-view qualification notice

The included GDS, LEF, and Liberty files are provisional interface-alignment views. Regenerate and sign off an implemented AHB macro with synthesis, place-and-route, DRC/LVS, extraction, and timing/power characterization before fabrication.

AHB signal reference

A transfer is active when HSEL=1, HTRANS[1]=1, and upstream HREADY=1. The model supports 32-bit word accesses at the configured data-register address.

Signal	Dir	Width	Description
HCLK	In	1	AHB clock. Address/control and data phases are sampled on rising edges.
HRESETn	In	1	Active-low asynchronous interface reset.
HSEL	In	1	Subordinate select from the address decoder.
HADDR	In	32	Byte address. Default model match is 0xE000_0004.
HTRANS	In	2	Transfer type. HTRANS[1]=1 identifies NONSEQ or SEQ active transfers.
HWRITE	In	1	1 for write; 0 for read.
HSIZE	In	3	Transfer size. Use 3'b010 for the supported 32-bit word access.
HBURST	In	3	Burst type. Accepted for bus compatibility; use single word transfers.
HPROT	In	4	Protection attributes. Accepted for integration compatibility.
HMASTER	In	4	Master identifier. Accepted for integration compatibility.
HMASTLOCK	In	1	Locked-transfer qualifier. Accepted for integration compatibility.
HWDATA	In	32	Write data, sampled in the AHB write data phase.
HREADY	In	1	Upstream transfer-ready input. A new address phase is accepted only when high.
HRDATA	Out	32	Read result word or empty-read sentinel.
HREADYOUT	Out	1	Transfer completion. Low inserts wait states.
HRESP	Out	2	2'b00 OKAY; 2'b01 ERROR for unsupported address, size, or transfer conditions.

Physical macro, scan, and analog pins

These pins are retained in the provisional physical views. Their electrical limits and production test sequence must be taken from the final characterized macro release.

Signal	Dir	Description
user_clk	In	Implementation clock used by the physical macro control domain.
user_rst	In	Implementation reset used by the physical macro control domain.
TM	In	Test-mode enable.
ScanInCC	In	Scan input for the column-control chain.
ScanInDL	In	Scan input for the left data chain.
ScanInDR	In	Scan input for the right data chain.
ScanOutCC	Out	Scan output for the column-control chain.
VDDC1, VDDC2	Power	Digital core supplies.
VDDA1, VDDA2	Power	Analog supplies.
VSS	Ground	Common ground.
Vcc_wl_read	Analog	Read word-line bias supply.
Vcc_wl_set	Analog	SET word-line bias supply.
Vcc_wl_reset	Analog	RESET word-line bias supply.
Vcc_set	Analog	SET bias supply.
Vcc_read	Analog	Read bias supply.
dc_bias	Analog	DC bias input.
Iref	Analog	Reference-current input.
Vbias	Analog	Analog bias-voltage input.
Vcomp	Analog	Comparator reference input.
Bias_comp2	Analog	Secondary comparator bias input.

Behavioral-only observability

The behavioral HDL also exposes scan/test observability ports named i_scan_si1, i_scan_se1, i_TM, o_TM, o_mux_sel, and 512-bit source-line, bit-line, and word-line state buses. These are simulation observability signals and are not a replacement for final hard-macro scan collateral.

Transaction and frame rules

Reset

Hold HRESETn low to reset the AHB interface, command state, and output queue. After release, the first three accepted writes are configuration words. They do not execute array commands.

Writes

Drive the address/control phase at 0xE000_0004 with HWRITE=1 and HSIZE=3'b010, then drive HWDATA in the following data phase. Hold the transfer stable while HREADYOUT is low.

Bits	Field	Definition
31:30	Mode	00 RESET, 01 READ, 10 COMPUTE, 11 SET
29:21	Row	9-bit global row address, 0-511
20:12	Column	9-bit global column address, 0-511
11	Status request	Return FSM error/status on the associated read
10	Compute all	Return all 32 columns in the selected 32-column bank
9	Compute start	Final COMPUTE packet and launch marker
8	Reserved	Write zero
7:0	PWM	Pulse-width/input activation value

Reads

Drive HWRITE=0. HREADYOUT remains low until a queued result is available. HRDATA and HRESP are valid when HREADYOUT returns high.

Bits	Field	Definition
31:25	Reserved	Zero
24:23	Mode	Operation that produced the result
22:14	Column	Global minuend column address
13:8	Coarse	Upper six bits of the signed 14-bit differential count
7:0	Fine	Lower eight bits of the signed 14-bit differential count

Differential signed-column convention

Physical columns are paired as (2*n, 2*n+1). Result 2*n is column[2*n] - column[2*n+1], and result 2*n+1 is the opposite difference. Result bits [22:14] identify the minuend column; bits [13:0] contain the signed 14-bit two's-complement difference.

Compute-all bank selection

When command bit 10 is set, the command column selects one 32-column bank. For example, column 68 selects columns 64 through 95. The model queues exactly 32 results with global column addresses.